Data Analysis Final Exam

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April 2026

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1 Time-Series

1.1 Introduction to Time Series

1.1.1 Data Mining and Sequence Data

What is Data Mining

• Data mining is the science and practice of discovering meaningful patterns, trends and knowledge from large and complex datasets.
• It integrates techniques from statistics, machine learning, databases and pattern recognition to support data-driven decision making.
• Rapid growth of digital data has made efficient and accurate data mining essential.
• Data mining is widely used in domains such as finance, healthcare, biology and web analytics.

Explanation:

ببساطة، Data mining يعني استخراج معلومات مفيدة من بيانات ضخمة جدًا.

بدل ما نبص على البيانات الخام، بنحاول نكتشف patterns واتجاهات تساعدنا نفهم البيانات.

المجال ده بيجمع بين statistics للتحليل، و machine learning للتوقع، و databases لتخزين البيانات.

ومع زيادة البيانات بشكل كبير حالياً (Big Data)، بقى مهم جدًا في تطبيقات الحياة الواقعية.

Sequence Data and Its Importance

• Sequence data is a key data type where the order of events is significant.
• Sequence data analysis reveals temporal, logical and structural patterns not visible in unordered data.

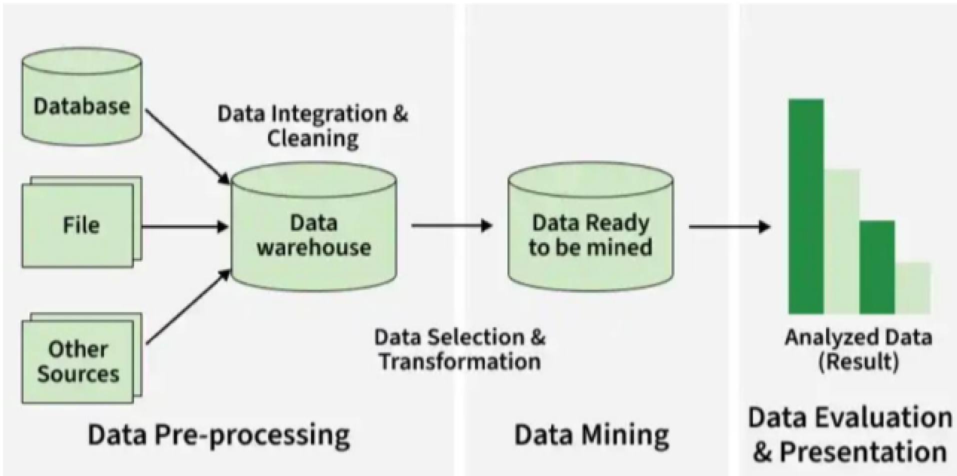
Explanation:

Sequence data بيعتمد بشكل أساسي على الترتيب.

لو غيرنا الترتيب، المعنى ممكن يتغير تمامًا.

أمثلة: stock prices user clicks sensor readings.

تحليل النوع ده بيساعدنا نكتشف patterns مع الوقت مش هتظهر لو البيانات كانت عشوائية.



Ordered Lists of Events or Elements

A sequence is an ordered list of events or elements.

- The order reflects time, position or logical dependency between elements.

Explanation:

ال sequence يعني بيانات مرتبة بترتيب معين.

الترتيب ده ممكن يكون:

- time → زي درجة الحرارة اليومية
- position → زي ترتيب الكلمات في الجملة
- logical dependency → زي خطوات حل مسألة أو process

1.1.2 Time-Series Data

Definition and Characteristics

- Time-series data consists of numerical values recorded at regular time intervals. Each observation is associated with a specific timestamp, making time an intrinsic component of the data.

Explanation:

Time-series data هو بيانات بتتجمع مع الزمن (كل ثانية، ساعة، يوم...).

كل قيمة بيكون ليها timestamp يعني وقت محدد، ف الزمن جزء أساسي من البيانات.

Numeric and Continuous Nature

- Numeric and continuous in nature

Explanation:
غالباً البيانات دي بتكون أرقام مش فئات (categories).
وكمان بتتغير بشكل تدريجي (smooth).

Time Dependency

- Observations are time-dependent

Explanation:

القيمة الحالية بتعتمد على القيم اللي قبلها.

مثال: سعر السهم النهارده بيعتمد على سعره امبارح.

Common Patterns in Time-Series Data

- Often exhibit trends, seasonality and cyclic patterns

Explanation:
البيانات دي غالباً بيكون فيها patterns بتتكرر أو بتتغير مع الزمن.
ودي مهمة جداً علشان نحلل البيانات ونتوقع المستقبل.

1.1.3 Time-Series Analysis and Forecasting

- Time-series analysis focuses on identifying underlying patterns such as:

Trend

- Trend: Long-term increase or decrease in data

Explanation:

الـ Trend بيوضح الاتجاه العام للبيانات (طالعة ولا نازلة).

مثال: زيادة المبيعات عبر السنين.

Seasonality

- Seasonality: Repeating patterns over fixed periods

  Explanation:

  Seasonality يعني pattern بيتكرر في فترات ثابتة.

  مثال: زيادة المبيعات في رمضان أو الصيف.

Cyclic Movements

- Cyclic movements: Fluctuations over irregular periods

  Explanation:

  شبه الـ trend لكن بدون وقت ثابت للتكرار. الـ Cycles

  (مثال: الدورات الاقتصادية boom و recession).

  - Time-series forecasting uses historical data to predict future values.

  Explanation:

بنستخدم البيانات القديمة علشان نتوقع اللي هيحصل في المستقبل.

Forecasting Models

Common models include:

- ARIMA (AutoRegressive Integrated Moving Average)
- SARIMA (Seasonal ARIMA)
- Long-memory and advanced machine learning models
- Subsequence matching and piecewise approximation techniques are widely used to efficiently analyze large time-series databases.

Explanation:

- ARIMA: موديل إحصائي أساسي للتوقع
- SARIMA: نفس ARIMA لكن بيأخد في الاعتبار الـ seasonality
- Long-memory models: بتفهم العلاقات طويلة المدى في البيانات
- Machine learning models: زي neural networks علشان توقعات أدق

1.1.4 Applications

- Stock market and financial analysis
- Weather and climate monitoring
- Medical signal analysis (ECG, EEG)
- Sensor and IoT data analysis

Explanation:

Time-series: موجود في تطبيقات كتير جداً

- Finance → توقع أسعار الأسهم
- Weather → توقع درجات الحرارة
- Medical → تحليل إشارات القلب والمخ
- IoT → تحليل بيانات الحساسات زي smart homes

1.2 Assessment Questions

Multiple Choice Questions (MCQs)

1. Data mining is mainly concerned with:
   (a) Storing raw data only
   (b) Discovering patterns and knowledge from large datasets
   (c) Deleting unnecessary data
   (d) Visualizing small datasets
2. Data mining integrates techniques from:
   (a) Biology only
   (b) Physics only
   (c) Statistics, machine learning, and databases
   (d) Networking only
3. Sequence data is important because:
   (a) Order does not matter
   (b) It only contains text
   (c) The order of events is significant
   (d) It cannot be analyzed
4. A sequence is defined as:

(a) Random data points
(b) An ordered list of events or elements
(c) Unstructured data
(d) Only numerical values

5. The order in a sequence may represent:
   (a) Color only
   (b) Size only
   (c) time, position, or logical dependency
   (d) Shape only

6. Time-series data consists of:
   (a) Images only
   (b) Text only
   (c) Numerical values recorded over time
   (d) Random categories

7. Time-series data is typically:
   (a) Categorical only
   (b) Numeric and continuous
   (c) Always binary
   (d) Static

8. Time dependency means:
   (a) Data is independent
   (b) Observations are time-dependent
   (c) Future values affect past values
   (d) Time has no role

9. Which of the following is a common time-series pattern?
   (a) Sorting
   (b) Searching
   (c) Trend
   (d) Indexing

10. Seasonality refers to:
    (a) Random fluctuations

(b) Repeating patterns over fixed periods
(c) Permanent increase
(d) Data cleaning

True & False Questions

1. Data mining is used to extract useful information from large datasets.
2. In sequence data, the order of elements is not important.
3. Time-series data always lacks timestamps.
4. Trend represents long-term increase or decrease in data.
5. Seasonality occurs at irregular time intervals.

Answers

MCQs:

1. b
2. c
3. c
4. b
5. c
6. c
7. b
8. b
9. c
10. b

True & False:

1. True
2. False
3. False
4. True
5. False

1.3 Symbolic Sequence Data

1.3.1 Definition of Symbolic Data

Ordered Events or Symbols

- Symbolic sequence data consists of ordered events or symbols, recorded with or without explicit time information. Unlike time-series data values are categorical rather than numeric.

  Explanation:

  ال Symbolic Data يعني البيانات مش أرقام، لكن رموز أو فئات.

  زي: منتجات، صفحات ويب، أو أفعال المستخدم (click, buy).

  كان ممكن يكون فيها وقت أو لأ، يعني الزمن مش شرط أساسي.

Categorical Rather Than Numeric Values

Explanation:

بدل ما يكون عندنا أرقام زي 10 و20، بيكون عندنا قيم فئوية زي (A, B, C).

هنا بنهتم بـ الترتيب مش القيمة العددية.

1.3.2 Examples of Symbolic Data

Customer Purchase Sequences

- Customer purchase sequences

  Explanation:

  مثال: عميل يشتري موبايل □ سماعات □ شاحن.

  الترتيب مهم لفهم سلوك العميل.

Web Clickstream Data

- Web clickstream data

  Explanation:

  تسلسل الصفحات اللي المستخدم بيزورها:

Home → Products → Cart → Checkout

وده بيساعد نفهم رحلة المستخدم.

User Navigation Paths

- User navigation paths

  Explanation:

  دي بتوضح المستخدم بيتحرك إزاي جوه النظام، وده مفيد لتحليل تجربة المستخدم.

1.3.3 Sequential Pattern Mining

Purpose of Sequential Pattern Mining

- Symbolic data is mainly analyzed using sequential pattern mining which aims to discover frequent subsequences occurring across multiple sequences.

  Explanation:

  هنا بنحاول نلاقي أنماط متكررة بين المستخدمين.

  مثلاً: ناس كتير بتشتري نفس التسلسل من المنتجات.

  The techniques used include:

Constraint-Based Pattern Matching

- Constraint-based pattern matching

  Explanation:

  بنحدد شروط معينة علشان نلاقي الأنماط (زي وقت معين أو ترتيب معين).

Apriori-Based Algorithms

- Apriori-based algorithms

  Explanation:

  ده خوارزمية مشهورة بتلاقي الأنماط المتكررة وبتشيل الأنماط النادرة تدريجياً.

Generalized Sequential Pattern (GSP) Algorithm

- Generalized Sequential Pattern (GSP) algorithm

  Explanation:

  ده تطوير لـ Apriori مخصوص علشان البيانات المتسلسلة.

1.3.4 Applications of Sequential Pattern Mining

These techniques help identify user behavior patterns, product associations and navigation trends, supporting applications such as recommendation systems and marketing analysis.

Explanation:

التقنيات دي بتساعدنا نفهم:

- سلوك المستخدم
- العلاقات بين المنتجات
- اتجاهات التصفح

Recommendation Systems

Explanation:

أنظمة بتقترح منتجات بناءً على سلوك المستخدم.

Marketing Analysis

Explanation:

بنستخدمها علشان نفهم اهتمامات العملاء ونحسن البيع.

User Behavior and Navigation Trends

Explanation:

تحليل طريقة استخدام المستخدم للنظام.

Tid	Time	Cid	Event(purchase product)
t1	11:45:30	c1	wheat, rice, fruit
t2	11:36:50	c2	rice, fruit
t1	12:00:01	c1	juice, rice
t2	01:00:34	c2	sugar, milk

1.4 Assessment Questions

Multiple Choice Questions (MCQs)

1. Symbolic sequence data consists of:
   (a) Random numerical values
   (b) Ordered events or symbols
   (c) Images only
   (d) Continuous signals

2. Symbolic sequence data may:
   (a) Always include time information
   (b) Never include time information
   (c) Be recorded with or without time information
   (d) Only include timestamps

3. The values in symbolic data are:
   (a) Numeric
   (b) Continuous
   (c) Categorical
   (d) Binary only

4. In symbolic data, the focus is mainly on:
   (a) Exact numerical values

(b) Order of elements
(c) Data storage
(d) Data compression

5. Which of the following is an example of symbolic data?
   (a) Temperature readings
   (b) Stock prices
   (c) Customer purchase sequence
   (d) Sensor voltage values

6. A customer purchase sequence helps in understanding:
   (a) Weather patterns
   (b) User behavior
   (c) Network speed
   (d) Data encryption

7. Web clickstream data represents:
   (a) Random browsing
   (b) Sequence of visited web pages
   (c) Numerical logs
   (d) Image processing steps

8. User navigation paths are useful for analyzing:
   (a) Hardware performance
   (b) User experience
   (c) Memory usage
   (d) CPU scheduling

9. Symbolic data is mainly analyzed using:
   (a) Regression analysis
   (b) Sequential pattern mining
   (c) Sorting algorithms
   (d) Data compression

10. Sequential pattern mining aims to discover:
    (a) Rare events
    (b) Frequent subsequences

(c) Random values
(d) Noise in data

11. Constraint-based pattern matching involves:
    (a) Ignoring all conditions
    (b) Applying specific conditions to find patterns
    (c) Sorting data randomly
    (d) Deleting data points

12. Apriori-based algorithms are used to:
    (a) Generate random sequences
    (b) Find frequent patterns and remove infrequent ones
    (c) Visualize data only
    (d) Encrypt data

13. The GSP algorithm is:
    (a) A sorting algorithm
    (b) A version of Apriori for sequential data
    (c) A database system
    (d) A visualization tool

14. Sequential pattern mining helps identify:
    (a) Hardware faults only
    (b) User behavior patterns and product associations
    (c) File sizes
    (d) Network protocols

15. One application of sequential pattern mining is:
    (a) Operating systems design
    (b) Recommendation systems
    (c) Compiler optimization
    (d) Image rendering

True & False Questions

1. Symbolic data always contains numerical values.
2. The order of elements is important in symbolic sequence data.
3. Symbolic data must always include time information.
4. Customer purchase sequences can reveal user behavior.
5. Sequential pattern mining is used to discover frequent subsequences.
6. Web clickstream data helps understand user navigation.
7. Apriori-based algorithms focus on rare patterns only.
8. GSP algorithm is designed for sequential data analysis. True !

Answers

MCQs:

1. b
2. c
3. c
4. b
5. c
6. b
7. b
8. b
9. b
10. b
11. b
12. b
13. b
14. b
15. b

True & False:

1. False
2. True
3. False
4. True
5. True
6. True
7. False
8. True

1.5 Biological Sequence Data

1.5.1 Definition of Biological Data

DNA Sequences

RNA Sequences

Protein Sequences

- Biological sequence data consists of DNA, RNA and protein sequences. These sequences are composed of nucleotides or amino acids arranged in a specific order, carrying essential biological information.

Explanation:

الـ Biological Data هو نوع من البيانات المتسلسلة لكن في مجال علم الأحياء.

DNA و RNA بيتكونوا من نيوكليوتيدات، والبروتينات من أحماض أمينية.

الترتيب هنا مهم جدًا لأنه بيحدد وظيفة الكائن أو البروتين.

1.5.2 Characteristics of Biological Sequences

Very Long and Complex Sequences

- Extremely long and complex sequences

Explanation:

السلاسل دي ممكن تكون طويلة جدًا (ملايين العناصر) ومعقدة.

High Dimensionality

- High dimensionality

  Explanation:

  فيها عدد كبير من الخصائص، وده بيصعب عملية التحليل.

Hidden Functional and Structural Patterns

- Contain hidden functional and structural patterns

  Explanation:

  فيها أنماط مخفية لازم نكتشفها علشان نفهم الوظيفة الحيوية.

1.5.3 Techniques for Biological Sequence Analysis

Sequence Alignment and Indexing

- Sequence alignment and indexing

  Explanation:

  بنقارن بين السلاسل علشان نلاقي تشابهات بينها.

Substitution Matrices and Trees

- Substitution matrices and trees

  Explanation:

  دي أدوات بتستخدم لقياس مدى التشابه بين العناصر.

Probabilistic Models

- Probabilistic models

  Explanation:

  بنستخدم الاحتمالات علشان نمثل البيانات الحيوية.

BLAST

- BLAST (Basic Local Alignment Search Tool) is one of the most widely used tools for biological sequence analysis, enabling fast and efficient similarity searches in large biological database

Explanation:

BLAST أداة قوية بتساعدنا نعمل مقارنة سريعة بين السلاسل ونلاقي التشابه.

1.5.4 Applications

- Gene identification and annotation
- Protein structure and function analysis
- Disease diagnosis and drug discovery

Gene Identification and Annotation

Explanation:

تحديد الجينات وفهم وظيفتها.

Protein Structure and Function Analysis

Explanation:

فهم تركيب البروتين ووظيفته.

Disease Diagnosis and Drug Discovery

Explanation:

تستخدم في تشخيص الأمراض وتطوير الأدوية.

1.5.5 Advantages of Sequence Data Mining

Revealing Temporal and Structural Patterns

- Reveals temporal and structural patterns

Supporting Prediction and Decision Making

- Supports prediction and decision making

Applications in Finance, Healthcare, and Bioinformatics

- Enables discoveries in finance, healthcare and bioinformatics

Handling Complex and High-Dimensional Data

- Handles complex, ordered and high-dimensional data.

  Explanation:

  المجال ده مهم لأنه:

  - بيكشف أنماط معقدة
  - بيساعد في التوقع
  - بيستخدم في مجالات كتير
  - يقدر يتعامل مع بيانات كبيرة ومعقدة

1.6 Assessment Questions

Multiple Choice Questions (MCQs)

1. Biological sequence data consists of:
   (a) Images and text
   (b) DNA, RNA, and protein sequences
   (c) Only numerical data
   (d) Sensor readings

2. DNA and RNA sequences are composed of:
   (a) Amino acids
   (b) Nucleotides
   (c) Proteins
   (d) Enzymes

3. Protein sequences are composed of:
   (a) Nucleotides
   (b) Amino acids
   (c) Lipids
   (d) Sugars

4. The order of elements in biological sequences is important because:
   (a) It reduces storage size
   (b) It determines biological function
   (c) It simplifies computation
   (d) It removes redundancy
5. Biological sequences are typically:
   (a) Short and simple
   (b) Extremely long and complex
   (c) Always binary
   (d) Random and unordered
6. High dimensionality in biological data means:
   (a) Few features
   (b) Large number of features
   (c) Small datasets
   (d) Simple structures
7. Biological sequences often contain:
   (a) Only visible patterns
   (b) Hidden functional and structural patterns
   (c) No patterns
   (d) Only numerical values
8. Sequence alignment is used to:
   (a) Compress data
   (b) Find similarities between sequences
   (c) Delete redundant data
   (d) Encrypt sequences
9. Substitution matrices are used to:
   (a) Store sequences
   (b) Measure similarity between elements
   (c) Visualize data
   (d) Sort sequences
10. Probabilistic models are used to:

(a) Eliminate uncertainty
(b) Represent data using probability
(c) Store large datasets
(d) Visualize graphs

11. BLAST is mainly used for:

    (a) Data compression
    (b) Fast similarity search in biological databases
    (c) Image processing
    (d) Network analysis

12. Gene identification and annotation involves:

    (a) Deleting genes
    (b) Identifying genes and understanding their function
    (c) Storing sequences only
    (d) Encrypting biological data

13. Protein structure analysis helps in:

    (a) Understanding protein function
    (b) Reducing data size
    (c) Sorting sequences
    (d) Ignoring patterns

14. Biological data analysis is used in:

    (a) Disease diagnosis and drug discovery
    (b) Only gaming
    (c) Only networking
    (d) Only graphics design

15. One advantage of sequence data mining is:

    (a) Ignoring patterns
    (b) Revealing temporal and structural patterns
    (c) Deleting data
    (d) Limiting applications

16. Sequence data mining supports:

    (a) Random guessing

(b) Prediction and decision making
(c) Data deletion
(d) Hardware optimization only

17. Sequence data mining is applied in:
    (a) Finance, healthcare, and bioinformatics
    (b) Only education
    (c) Only sports
    (d) Only entertainment

18. Sequence data mining can handle:
    (a) Simple and small data only
    (b) Complex and high-dimensional data
    (c) Only text data
    (d) Only images

True & False Questions

1. Biological sequence data includes DNA, RNA, and proteins. ✓
2. DNA sequences are made of amino acids. ✗
3. Protein sequences are made of amino acids. ✓
4. The order of elements in biological sequences is not important. ✗
5. Biological sequences are usually short and simple. ✗
6. Biological data often contains hidden patterns. ✓
7. Sequence alignment helps find similarities between sequences. ✓
8. BLAST is a slow and inefficient tool. ✗
9. Probabilistic models use probability to represent data. ✓
10. Sequence data mining helps in prediction and decision making. ✓

Answers

MCQs:

1. b
2. b
3. b
4. b
5. b
6. b
7. b
8. b
9. b
10. b
11. b
12. b
13. a
14. a
15. b
16. b
17. a
18. b

True & False:

1. True
2. False
3. True
4. False
5. False
6. True

7. True
8. False
9. True
10. True

1.7 Basic DateTime Operations in Python

1.7.1 Overview of the DateTime Module

Purpose of the Module

• Python has an in-built module named DateTime to deal with dates and times in numerous ways. In this article, we are going to see basic DateTime operations in Python.

Explanation:

Python provides a ready module to handle date and time بسهولة بدون ما نكتب كل حاجة من الصفر.

Basic Date and Time Handling in Python

Explanation:
Using this module, we can create, format, compare, and manipulate dates and times.

1.7.2 Main Classes in the datetime Module

There are six main object classes with their respective components in the datetime module mentioned below:

- datetime.date
- datetime.time
- datetime.datetime
- datetime.tzinfo
- datetime.timedelta
- datetime.timezone

• Now we will see the program for each of the functions under datetime module mentioned above.

Explanation:
Each class is responsible for a specific part:

- date → التاريخ فقط
- time → الوقت فقط
- datetime → الاثنين مع بعض
- timedelta → الفرق بين وقتين

1.7.3 datetime.date()

We can generate date objects from the date class. A date object represents a date having a year, month and day.

Syntax
datetime.date(year, month, day)

strftime to print day, month and year in various formats. Here are some of them are:

- current.strftime("%m/%d/%y")
- current.strftime("%b-%d-%Y")
- current.strftime("%d/%m/%Y")
- current.strftime("%B %d, %Y")

Explanation:
strftime is used to format date in different shapes.
Example: month/day/year or full month name.

1.7.4 datetime.time()

A time object generated from the time class represents the local time.

Components:

- hour
- minute
- second
- microsecond
- tzinfo

Syntax
datetime.time(hour, minute, second, microsecond)

1.7.5 datetime.datetime()

datetime.datetime() module shows the combination of a date and a time.

Components:

- year
- month
- day
- hour
- minute
- second
- microsecond
- tzinfo

Syntax
datetime.datetime(year, month, day)

datetime.datetime(year, month, day, hour, minute, second, microsecond)

Current date and time using the strftime() method in different ways:

- strftime("%d")
- strftime("%m")
- strftime("%Y")
- strftime("%H:%M:%S")
- strftime("%m/%d/%Y, %H:%M:%S")

Explanation:
datetime is the most important class لأنه يجمع التاريخ والوقت مع بعض.

1.7.6 datetime.timedelta()

Definition and Meaning

It shows a duration that expresses the difference between two date, time or datetime instances to microsecond resolution.

Explanation:
Used to calculate الفرق بين تاريخين أو وقتين.

Past and Future Day Calculations

- Here we implemented some basic functions and printed past and future days.

Minimum and Maximum Values

- Also, we will print some other attributes of timedelta max, min that show maximum days and time, minimum date and time

Arithmetic Operations on Dates and Times

- Here we will also apply some arithmetic operations on two different dates and times.

1.7.7 Time Module Functions

time() method

The time() method returns the time as a floating-point number expressed in seconds since the epoch, in UTC

Syntax
time.time([])

ctime() method

ctime() method converts a time expressed in seconds since the epoch to a string representing local time.

Syntax
time.ctime([sec])

sleep() method

Python time method sleep() stops execution for the given number of seconds.

Syntax
time.sleep([sec])

localtime() method

localtime() method converts number of seconds to local time.

Syntax
time.localtime([sec])

gmtime() method

gmtime() method converts a time expressed in seconds since the Epoch to a struct_time in UTC.

Syntax

time.gmtime([sec])

mktime() method

It is the inverse function of localtime() method.

Syntax

time.mktime([t])

asctime() method

Python time method asctime() converts a struct_time representing a time to a 24-character string.

Syntax

time.asctime([t])

1.8 Assessment Questions

Multiple Choice Questions (MCQs)

1. The DateTime module in Python is used to:

   (a) Handle images

   (b) Work with dates and times

   (c) Process text only

   (d) Manage files

2. Using the datetime module, we can:

   (a) Only store dates

   (b) Create, format, compare, and manipulate dates and times

   (c) Only delete dates

   (d) Only print time

3. Which of the following is NOT a class in the datetime module?

   (a) datetime.date

   (b) datetime.time

   (c) datetime.random

(d) datetime.timedelta

4. The datetime.date class represents:

   (a) Time only
   (b) Date with year, month, and day
   (c) Both date and time
   (d) Time zone only

5. The correct syntax to create a date object is:

   (a) datetime.date(day, month, year)
   (b) datetime.date(year, month, day)
   (c) datetime.date(month, year, day)
   (d) datetime.date(year, day)

6. The strftime() function is used to:

   (a) Compare dates
   (b) Format date and time
   (c) Delete time
   (d) Add dates

7. A time object represents:

   (a) Date only
   (b) Local time
   (c) Global time only
   (d) Year only

8. Which of the following is NOT a component of time?

   (a) Hour
   (b) Minute
   (c) Month
   (d) Second

9. datetime.datetime represents:

   (a) Only date
   (b) Only time
   (c) Combination of date and time
   (d) Only timezone

10. Which class is used to calculate the difference between two dates?
    (a) datetime.time
    (b) datetime.timedelta
    (c) datetime.date
    (d) datetime.tzinfo
11. timedelta is used to:
    (a) Store text
    (b) Represent duration between dates or times
    (c) Display images
    (d) Sort data
12. The time() function returns:
    (a) Current date
    (b) Time in seconds since epoch
    (c) Formatted string
    (d) Local time only
13. The ctime() function is used to:
    (a) Convert seconds to string time
    (b) Pause execution
    (c) Return UTC time
    (d) Convert string to seconds
14. The sleep() function is used to:
    (a) Stop execution for a given number of seconds
    (b) Convert time
    (c) Format time
    (d) Get system time
15. localtime() converts:
    (a) String to time
    (b) Seconds to local time
    (c) UTC to seconds
    (d) Time to string
16. gmtime() returns:

    (a) Local time
    (b) UTC time
    (c) String format
    (d) Random time

17. mktime() is:
    (a) Same as gmtime()
    (b) Inverse of localtime()
    (c) Used for formatting
    (d) Used for delay

18. asctime() converts:
    (a) Seconds to integer
    (b) struct_time to string
    (c) String to struct_time
    (d) Time to float

True & False Questions

1. The datetime module is built into Python. ✓
2. datetime.date includes time information. ✗
3. strftime() is used for formatting date and time. ✓
4. datetime.time includes year and month. ✗
5. datetime.datetime combines date and time. ✓
6. timedelta represents the difference between two dates or times. ✓
7. time.time() returns time in seconds since the epoch. ✓
8. sleep() function speeds up execution. ✗
9. gmtime() returns UTC time. ✗ True!
10. mktime() is the inverse of localtime(). ✓

Answers

MCQs:

1. b
2. b
3. c
4. b
5. b
6. b
7. b
8. c
9. c
10. b
11. b
12. b
13. a
14. a
15. b
16. b
17. b
18. b

True & False:

1. True
2. False
3. True
4. False
5. True
6. True

7. True
8. False
9. True
10. True

1.9 Time Series Analysis and Visualization in Python

1.9.1 Introduction to Time Series Analysis

Definition of Time Series Data

• Time series data is information collected in sequence over time. It shows how things change at different points, like stock prices every day or temperature every hour.

Explanation:
Time series = data ordered by time.
Example: daily sales, hourly temperature.

Importance in Real-World Applications

• It is used in industries such as finance, pharmaceuticals, social media, and research.

Explanation:
Used in many fields because most real data depends on time.

Goal of Analysis and Visualization

• Analyzing and visualizing this data helps us find trends, seasonal patterns, and behaviors.
• These insights support forecasting and guide better decision-making.
• The main goal is to study data in time order to extract meaningful patterns and predictions.

Explanation:
We analyze time data to understand the past and predict the future.

1.9.2 Core Concepts in Time Series Analysis

Trend

• Trend: It represents the general direction in which a time series is moving over an extended period. It checks whether the values are increasing, decreasing or staying relatively constant.

Explanation:
Trend = overall direction (up / down / .(

Seasonality

- Seasonality: Seasonality refers to repetitive patterns or cycles that occur at regular intervals within a time series corresponding to specific time units like days, weeks, months or seasons.

Explanation:
Repeated patterns at fixed time (مثلاً كل سنة أو كل شهر).

Moving Average

- Moving average: It is used to smooth out short-term fluctuations and highlight longer-term trends or patterns in the data

Explanation:
Used to reduce noise and make trend clearer.

Noise

- Noise: It represents the irregular and unpredictable components in a time series that do not follow a pattern.

Explanation:
Random changes in data (no clear reason).

Differencing

- Differencing: It is used to make the difference in values of a specified interval. By default it's 1 but we can specify different values for plots.

Explanation:
We subtract consecutive values to remove trend and make data stable.

Stationarity

- Stationarity: A stationary time series is statistical properties such as mean, variance and autocorrelation remain constant over time.

Explanation:
Important concept: data should not change its behavior over time.

Order of Differencing

- Order: The order of differencing refers to the number of times the time series data needs to be differenced to achieve stationarity.

Explanation:
How many times we apply differencing (1, 2, ...).

Autocorrelation

- Autocorrelation: Autocorrelation is a statistical method used in time series analysis to quantify the degree of similarity between a time series and a lagged version of itself.

Explanation:
Compare current value with previous values.

Resampling

- Resampling: Resampling is a technique in time series analysis that is used for changing the frequency of the data observations.

Explanation:
Example: convert daily data → monthly data.

1.9.3 Types of Time Series Data

- Time series data can be classified into two sections:

Continuous Time Series

- Continuous Time Series: Data recorded at regular intervals with a continuous range of values like temperature, stock prices, Sensor Data, etc.

Explanation:
Values are numeric and continuous.

Discrete Time Series

- Discrete Time Series: Data with distinct values or categories recorded at specific time points like counts of events, categorical statuses, etc.

Explanation:
Values are counts or categories.

1.9.4 Visualization Approaches

Line Plots and Area Charts

- Use line plots or area charts for continuous data to highlight trends and fluctuations.

Explanation:
Best for showing changes over time.

Bar Charts and Histograms

- Use bar charts or histograms for discrete data to show frequency or distribution across categories.

Explanation:
Used for counts and comparisons.

1.10 Assessment Questions

Multiple Choice Questions (MCQs)

1. Time series data is defined as:
   (a) Random data
   (b) Data collected over time in sequence
   (c) Only categorical data
   (d) Static data
2. An example of time series data is:
   (a) Student names
   (b) Daily stock prices
   (c) Colors list
   (d) File sizes
3. Time series analysis is widely used in:
   (a) Finance and research
   (b) Only gaming
   (c) Only networking
   (d) Only graphics
4. The goal of time series analysis is to:
   (a) Delete data
   (b) Find patterns and make predictions
   (c) Store data only
   (d) Encrypt data
5. Trend represents:

(a) Random noise
(b) General direction of data over time
(c) Data storage
(d) Data format

6. Seasonality refers to:
   (a) Random variation
   (b) Repeating patterns at fixed intervals
   (c) Increasing trend only
   (d) Decreasing trend only

7. Moving average is used to:
   (a) Increase noise
   (b) Smooth short-term fluctuations
   (c) Delete data
   (d) Store values

8. Noise in time series represents:
   (a) Regular patterns
   (b) Predictable behavior
   (c) Irregular and random variations
   (d) Trends

9. Differencing is used to:
   (a) Add values
   (b) Subtract consecutive values
   (c) Multiply values
   (d) Sort data

10. Stationarity means:
    (a) Data changes randomly
    (b) Statistical properties remain constant over time
    (c) Data is deleted
    (d) Data is sorted

11. Order of differencing refers to:
    (a) Number of variables

(b) Number of times differencing is applied
(c) Size of dataset
(d) Type of data

12. Autocorrelation measures:

    (a) Difference between datasets
    (b) Similarity between a series and its lagged values
    (c) Data size
    (d) Frequency of data

13. Resampling is used to:

    (a) Delete data
    (b) Change frequency of observations
    (c) Sort data
    (d) Compress files

14. Continuous time series data is:

    (a) Categorical only
    (b) Numeric and continuous values
    (c) Random data
    (d) Text data

15. Discrete time series data is:

    (a) Continuous values
    (b) Counts or categories
    (c) Always numeric
    (d) Always random

16. Line plots are best used for:

    (a) Discrete data only
    (b) Continuous data over time
    (c) Text analysis
    (d) File comparison

17. Bar charts are useful for:

    (a) Continuous trends
    (b) Frequency of categories

(c) Time smoothing
(d) Noise reduction

True & False Questions

1. Time series data is ordered by time.
2. Time series analysis is not useful in real-world applications.
3. Trend shows the overall direction of data.
4. Seasonality occurs at irregular intervals.
5. Moving average helps smooth data.
6. Noise represents predictable patterns.
7. Differencing helps make data more stable.
8. Stationarity means statistical properties change over time.
9. Autocorrelation compares current and past values.
10. Resampling changes the frequency of data.

Answers

MCQs:

1. b
2. b
3. a
4. b
5. b
6. b
7. b
8. c
9. b

10. b
11. b
12. b
13. b
14. b
15. b
16. b
17. b

True & False:

1. True
2. False
3. True
4. False
5. True
6. False
7. True
8. False
9. True
10. True